SAINT JOSEPH'S UNIVERSITY
Erivan K. Haub School of Business

DSS 455: Machine Learning for Business Applications II
Draft Syllabus ‡

Instructor:	Liyuan Liu, Ph.D.
Email:	lliu@sju.edu
Tel. No.:	610-660-1635
Class Meetings	TR: 2-3PM

Office No.:	Mandeville Hall 349
Office Hours	Tuesdays: 12:20pm-1:50pm, 3pm-7:30pm To help me manage time and make sure everyone gets a chance to meet, please use the link below to book a slot: https://calendly.com/lliu-sju/30min

COURSE DESCRIPTION

The course is a follow-on to DSS 451 (Introduction to AI and ML Business Applications). A successful professional in the field of Business Intelligence & Analytics (BI&A) is required to be conversant (or at least familiar) with the new era of AI and ML to aid better business decisions.

The course will build on the methods learned in DSS 451 and will supplement the student's knowledge with some of the most commonly used Machine Learning Algorithms today. This will include Naïve Bayes, Natural Language Processing and Neural Networks and Deep Learning, which are one of the fastest growing and widely used ML algorithms in the industry. Python will be the primary mode of deploying and analyzing Machine Learning Models in class. Frequent use of real-world business case studies will be made in order to help connect these concepts to business applications. Through a capstone project, students will also be able to learn to apply these ML methods to problems in an organization and have a framework to navigate ethical and societal discussions surrounding AI.

This course WILL NOT teach you how to code a programming language. Neither will the focus be on the math behind the algorithms – but instead, we will focus on the applications, analysis and interpretation of machine learning methods. Please be proficient with Python at a level commensurate with successful completion of DSS 325

LEARNING OBJECTIVES

- Continue adding to your skill sets with additional ML algorithms
- Understand how machines make decisions using algorithms. And which ML algorithm may be appropriate for what situation.
- Using Python to implement ML algorithms
- Improve communication between business owners and data scientists/analysts

PLANNED COURSE SCHEDULE ‡

Week #	Date (mm/dd)	Topics	Important Due Dates (All due time is 11:59pm)
1	01/12-01/16	• Course Introduction and Logistics. • Review of DSS 451 (Prev. Course) • Clustering: K-means	
2	01/19-01/23	• Variable Selections: PCA, Variable Clustering	01/23: Assignment 1: K-means
3	01/26-01/30	• Variable Selection Practice • Association Rule	
4	02/02-02/06	• Assignment 1 Presentation • Hands-on Project	02/06: Assignment 2: Variable Selection
5	02/09-02/13	• Assignment 2 Presentation • Optimization • Feedforward Artificial Neural Network I	
6	02/16-02/20	• Feedforward Artificial Neural Network II	02/20: Assign Mid-term Take Home Exam
7	02/23-02/27	• Convolutional Neural Networks (CNN)	02/28: Mid-term Take Home Exam Due
8	03/02-03/06	Spring Break No Class	
9	03/09-03/13	• Image Classification Using CNN Hands on Project	03/14: Final Project Proposal Due
10	03/16-03/20	• Mid-term Exam Presentation • Recurrent Neural Networks (RNN)	03/20: Assignment 3: CNN Image Classification
11	03/23-03/27	• Long-short Term Memory (LSTM, GRU)	
12	03/30-04/03	• Natural Language Processing Basics and Generative AI	04/03: Assignment 4: RNN and LSTM
13	04/06-04/10	• Assignment 4 Presentation • Generative AI (Transformer: BERT, BERT variations, GPT, etc)	
14	04/13-04/17	• Natural Language Processing Hands-on Project	

13	04/20-04/24	• Natural Language Processing Hands-on Project	
14	04/27-04/30	Final Project Prepareand Final Project Presentation	

‡ *Contents and dates subject to change. Guest Speakers may be featured during the term, depending on availability and appropriateness to the topics being discussed.*

Electronic Equipment Use

You will use the Python software for the analysis and interpretation of the datasets used in this class. About 50% of each week (after Week 2) will be spent working hands-on with Python/Scikit Library applying the algorithms and interpreting the analyses.

Use of Canvas and University email
Canvas / SJU email will be the preferred method of communication for class-related announcements, events, changes, etc. It is the student's responsibility to check Canvas and his/her University email account regularly. Instructor is not responsible for using email accounts other than the student's assigned University account.

COURSE POLICIES

Pre-Requisites

- DSS 451 (ML course #1) – No exceptions
- DSS 325

Required Course Materials

- Instructor will provide slides, datasets and notes required for the course
- Software: Python

Optional but recommended texts (in order of importance):

- Deep Learning by Yoshua Bengio and Aaron Courville
- Machine Learning For Dummies by John Paul Mueller, Luca Massaron
- Introduction to Machine Learning with Python: A Guide for Data Scientists, Müller & Guido
- Machine Learning For Dummies®, IBM Limited Edition (Free?)
  - Link: https://www.ibm.com/downloads/cas/GB8ZMQZ3
- The Master Algorithm: How the Quest for the Ultimate Learning Machine Will Remake Our World by Pedro Domingos – Read for pleasure

If a bit of Math doesn't scare you

- The hundred-page machine learning book (http://themlbook.com/) – Pay what you want
- An Introduction to Statistical Learning (with applications in R) Free (www.statlearning.com)

- Other readings as assigned by instructor and/or posted on the course schedule and Canvas site.

Students with Disabilities
Reasonable academic accommodations may be provided to students who submit appropriate

documentation of their disability. Students are encouraged to contact Dr. Christine Mecke in the Office of Student Disability Services, Bellarmine, B-10, at cmecke@sju.edu; or at 610.660.1774 (voice), or 610.660.1620 (TTY), for assistance with this issue. The university also provides an appeal/grievance procedure regarding requested or offered reasonable accommodations through Dr. Mecke's office. More information is available at www.sju.edu/sds

Make-up Policy

Students are expected to be present for all in-class quizzes, examinations as listed on the syllabus or announced by the instructor. Make-up exams require valid documentation (this includes medical and other family emergencies). Valid documentation of an absence must be requested through the Dean's Office and presented to the instructor in order to reschedule an exam. ***There will be no exceptions made to this policy.*** Please do not request makeup exams or extended assignment deadlines based on family leisure travel. Consult the University's Academic Calendar and plan accordingly. *Missed quizzes / exams will result in a grade of zero (0) points for that exam.*

RESPONSIBLE EMPLOYEE REPORTING

While I want you to feel comfortable coming to me with issues you may be struggling with or concerns you may be having, please be aware that I have some reporting requirements that are part of my job responsibilities at Saint Joseph's University. For example, if you inform me of an issue of sexual harassment, discrimination or sexual misconduct (Sexual Assault, Sexual Harassment, Sexual Exploitation, Domestic Violence, Dating Violence, or Stalking) I will keep the information as private as I can, but I am required to bring it to the attention of the institution's Title IX Coordinator, Mary-Elaine Perry. If you would like to speak with her directly, she can be reached at mperry01@sju.edu or 610-660-1145.

Counseling and Psychological Services (CAPS) can provide confidential support and can assist you with determining how you would like to address this issue. CAPS is located at 504A Merion Gardens and can be reached at 610-660-1090. They also have walk-in hours M-F 11- 12:30 in LaFarge. REPP, Rape Education and Prevention Program provides a 24 hour confidential helpline (610-733-9650) as well. For more information about your confidential options at SJU, as well as resources in the community go to the Support Website.

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Anonymous Student Feedback

The following link allows students to provide anonymous feedback directly to the Decision & System Sciences Department (DSS). Students who believe the situation they are experiencing warrants immediate attention can notify the Department via this anonymous survey. Students may also email the Department Chair (Dr. Virginia Miori) directly at vmiori@sju.edu). Please note that this link is not meant to be used as a faculty evaluation and does not replace the student evaluation at the end of the semester.

https://sju.co1.qualtrics.com/jfe/form/SV_4NSi1TCKdsHWF5X

COURSE GRADING

Final course grade will be assigned based on the following scale. Please note that *extra-credit work is* ***not*** *available at any point for this course since there are no weekly assignments*.

Python Assignments	35% (announced in Class)
Mid Term Exam	25%
Final Project / take-home exam	35% (Individual Project)
Class Participation	5%
TOTAL	**100 points**

Final Grade Distribution

A	95-100 points	C+	77-<80 points
A-	90 - <95 points	C	73-<77 points
B+	87-<90 points	C-	70-<73 points
B	83-<87 points	D	65-<70 points
B-	80-<83 points	F	0-< 65 points

Note: Points round up to the nearest 0.5. (Eg: 86.4 is a B, 86.6 is a B+)
Canvas provides a "dumb"/unweighted average, which you should ignore.
I am unable to delete that column from the Canvas Gradebook

Readings and References

- May be assigned on a weekly basis by the instructor.